

Kafka Offset Reset Problem SOP

Objective

Prevent consumers from reading unintended old or new messages due to offset reset or committed offset issues.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T027
Issue	Kafka offset reset problem
Category	Streaming
Service	Kafka
Severity	Medium
Status	Resolved
Description	Consumer reading old messages
Expected Resolution	Set auto.offset.reset to latest

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L019	5/1/2026 10:09	ERROR	Offset reset issue	KAFKA_OFFSET
L031	5/1/2026 10:15	ERROR	Offset reset issue	KAFKA_OFFSET

Procedure

1. Confirm whether consumer is replaying old messages or skipping messages.
2. Identify consumer group, topic, partition, current offset, and committed offset.
3. Check auto.offset.reset value and whether committed offsets exist.
4. Validate latest and earliest offsets using kafka-consumer-groups.
5. Stop affected consumers before changing offsets.
6. Reset offsets to latest, earliest, timestamp, or specific offset as approved.
7. Restart consumers and verify processing begins from expected position.
8. Document before and after offsets in the incident ticket.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
```

```
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Old messages replayed	Offset reset to earliest	Set or reset offset to latest
Messages skipped	Offset moved too far ahead	Reset to timestamp or specific offset
Offset reset ignored	Committed offsets already exist	Reset committed offsets after stopping consumer
Repeated replay	Consumer group changed	Validate group ID configuration

Best Practices

1. Never reset production offsets without approval.
2. Record offsets before changes.
3. Use separate consumer groups for testing.
4. Keep group IDs stable across deployments.

References

Apache Kafka Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | Kafka CLI Runbook